Revolutionising B.Tech

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ENVIRONMENT AND ECOSYSTEM

Module-1

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Aim

To equip students in the fundamentals and understanding of various components and importance of environments

To teach the students with the knowledge of various types of ecosystem, cycling of nutrients and climate regulation

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Objective

a. Explain the scope and importance of environmental sciences

b. Educate the structure and composition of various aspects of environmental components

c. Explain about various types of ecosystem and the relationship between the food chain and various organisms.

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Syllabus

Environment -definition - scope and importance of environmental science- structure and composition of atmosphere, hydrosphere, lithosphere and biosphere- Ecosystem -structure - biotic and abiotic components - functions - energy flow in ecosystems, food chain and food web- types of ecosystems - forest, aquatic and grass land ecosystems

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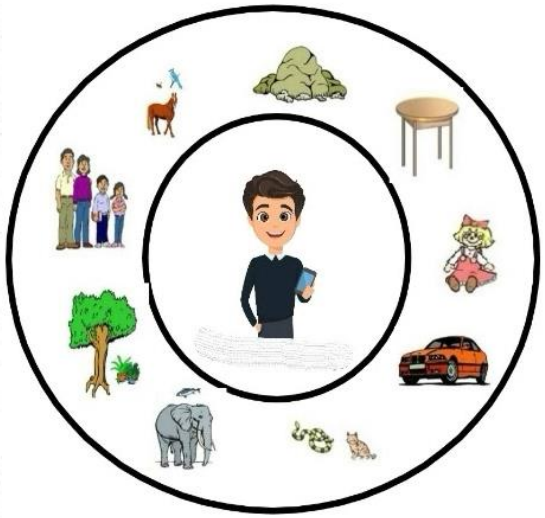
What is an Environment?

Environment can be defined as all surrounding conditions that have direct impact on all living and non-living things

Derived from French word “Environ” which means Encircle

Some examples of natural environments include rivers, mountains, forests and beaches.



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Environmental Sciences

- An interdisciplinary branch of science that investigates questions related to the human population, resources, and damages caused by pollution and disturbance.

- Environmental science is integrative and involves complex biology, chemistry, politics, sociology, geology, agriculture, economics.

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Scope of environmental studies

Environmental studies discipline has multiple and multilevel scopes.

- The study creates awareness among the people to know about various renewable and non-renewable resources of the region. The endowment or potential, patterns of utilization and the balance of various resources available for future use in the state of a country are analysed in the study.
- It provides the knowledge about ecological systems and cause and effect relationships
- It provides necessary information about biodiversity richness and the potential dangers to the species of plants, animals and microorganisms in the environment
- The study enables one to understand the causes and consequences due to natural and induced disasters (flood, earthquake, landslide, cyclones etc.,) and pollutions and measures to minimize the effects
- It enables one to evaluate alternative responses to environmental issues before deciding an alternative course of action

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- The study enables environmentally literate citizens (by knowing the environmental acts, rights, rules, legislations, etc.) to make appropriate judgments and decisions for the protection and improvement of the earth.

- The study exposes the problems of over population, health, hygiene, etc. and the role of arts, science and technology in eliminating/ minimizing the evils from the society

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Importance of environmental studies

To solve the global problems:

Environment Issues Being of International Importance It has been well recognised that environment issues like global warming and ozone depletion, acid rain, marine pollution and biodiversity are not merely national issues but are global issues and hence must be tackled with international efforts and cooperation.

To reduce the impact of developments:

Developmental activities such as, Urbanization, Industrial Growth, Transportation Systems, Agriculture and Housing etc. have been led to numerous environmental issues. In order to reduce/tackle the impact of such developments on environment it is important to study about environmental science.

To find a better way of Living:

Environmental science is more concerned with discovering ways to live more sustainably. This means utilizing present resources in a manner that conserves their supplies for the future.

Efficient Utilization of Natural Resources:

Environmental science teaches us to use natural resources efficiently by:

Appropriately putting into practice environmental conservation methods
Using the right techniques to minimize exploitation
Thorough training of human resources

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To focus more on environmental sustainability:

Environmental science aims to provide better understanding on the increasing population risk, overuse of fossil fuels, conservation of biodiversity.

To create awareness environmental problems at local, national, international levels:

Environmental problems at local, national, and international levels mostly occur due to lack of awareness. Environmental science aims to educate and equip learners with the necessary environmental skills

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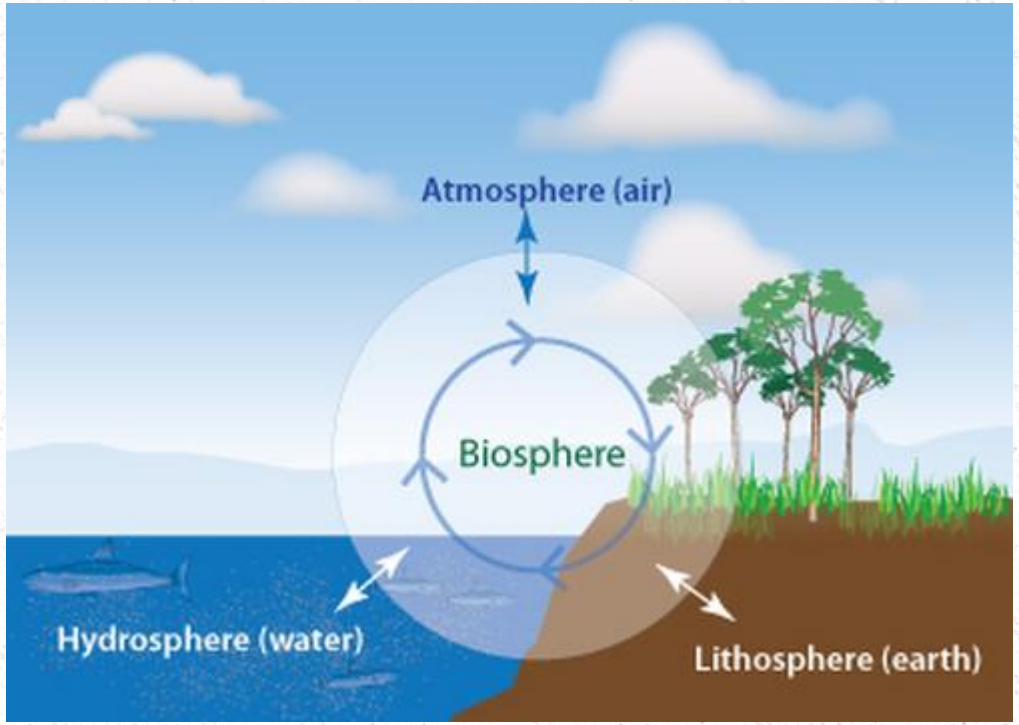
Structure and Composition of Atmosphere, Hydrosphere, Lithosphere and Biosphere

- **Litho**sphere - litho referring to rocks and minerals
- **Hydro**sphere - hydro referring to water
- **Bio**sphere - bio referring to life
- **Atmo**sphere - atmo referring to steam and vapor



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Hydrosphere:

The Hydrosphere comprises all types of water resources oceans, seas, lakes, rivers, streams, reserviour, polar icecaps, glaciers, and ground water.

(i) Nature 97% of the earth's water supply is in the oceans,

(ii) About 2% of the water resources is locked in the polar icecaps and glaciers.

(iii) Only about 1% is available as fresh surface water-rivers, lakes streams, and ground water fit to be used for human consumption and other uses.

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Atmosphere

The atmosphere implies the protective blanket of gases, surrounding the earth:

(a) It sustains life on the earth.

(b) It saves it from the hostile environment of outer space.

(c) It absorbs most of the cosmic rays from outer space and a major portion of the electromagnetic radiation from the sun.

(d) (d) It transmits only here ultraviolet, visible, near infrared radiation (300 to 2500 nm) and radio waves. (0.14 to 40 m) while filtering out tissue-damaging ultraviolet waves below about 300 nm.

(e) The atmosphere is composed of nitrogen and oxygen. Besides, argon, carbon dioxide, and trace gases.

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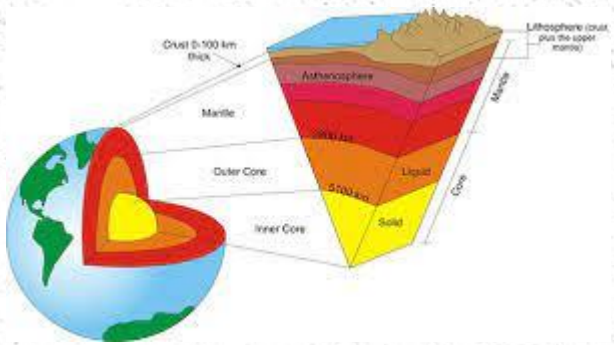
Lithosphere



The lithosphere includes a various number of different landforms such as mountains, valleys, rocks, minerals and soil. The lithosphere is constantly changing due to forces and pressures such as the sun, wind, ice, water and chemical changes.

The earth's surface is composed into two types of lithospheres. There are known as the oceanic and continental lithospheres.

The oceanic lithosphere includes the uppermost layers of mantle which is topped with a thin yet heavy oceanic crust. This is where the hydrosphere and lithosphere meet.

The continental lithosphere include the uppermost layers of mantle which is topped with a thick yet light continental crust. This is where the atmosphere, biosphere and hydrosphere meet the lithosphere.

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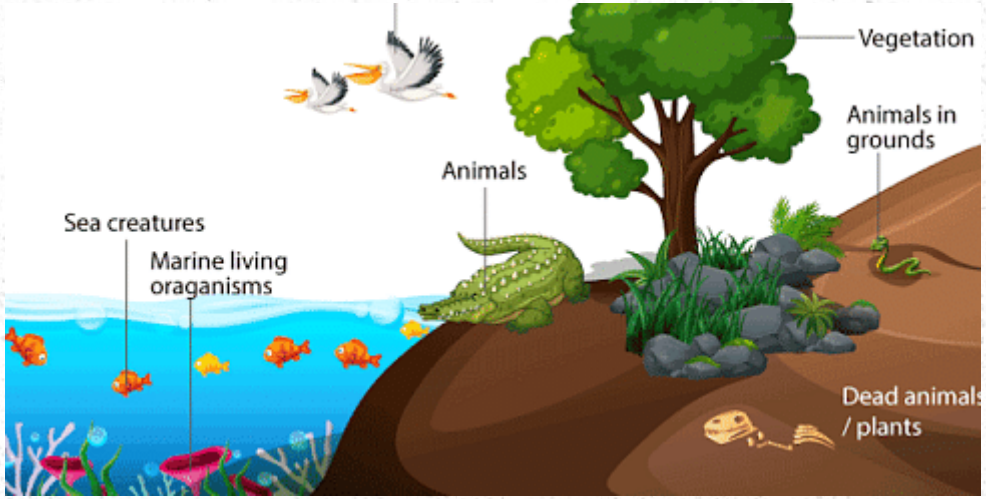
Biosphere

•The biosphere is defined as an area that comprises **all living species** as well as their **products**.

•The biosphere encompasses everything from tree roots to ocean trenches, lush rain forests, towering mountaintops, and transition zones like this one, where ocean and terrestrial ecosystems collide.

•The biosphere encompasses all three domains since life exists on the land, in the air, and in the water.

•Although the biosphere spans around 20 kilometers (12 miles) from top to bottom, practically all species may be found between 500 meters (1,640 feet) below sea level and 6 kilometers (3.75 miles) above sea level.



Components of Biosphere

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Ecosystem

Components of Ecosystem

- **Abiotic components**
  - **Physical factors**
    - Temperature
    - Humidity
    - Light
    - Atmospheric pressure
  - **Inorganic substances**
    - Water
    - Oxygen
    - Carbon dioxide
    - Nitrogen
  - **Organic substances**
    - Proteins
    - Carbohydrates
    - Lipids
- **Biotic components**
  - Producers (Green plants)
  - Consumers (Animals)
  - Decomposers (Microorganisms)

Ecosystem is defined as structural and functional unit of the biosphere, comprising living and non-living factors and their interaction.

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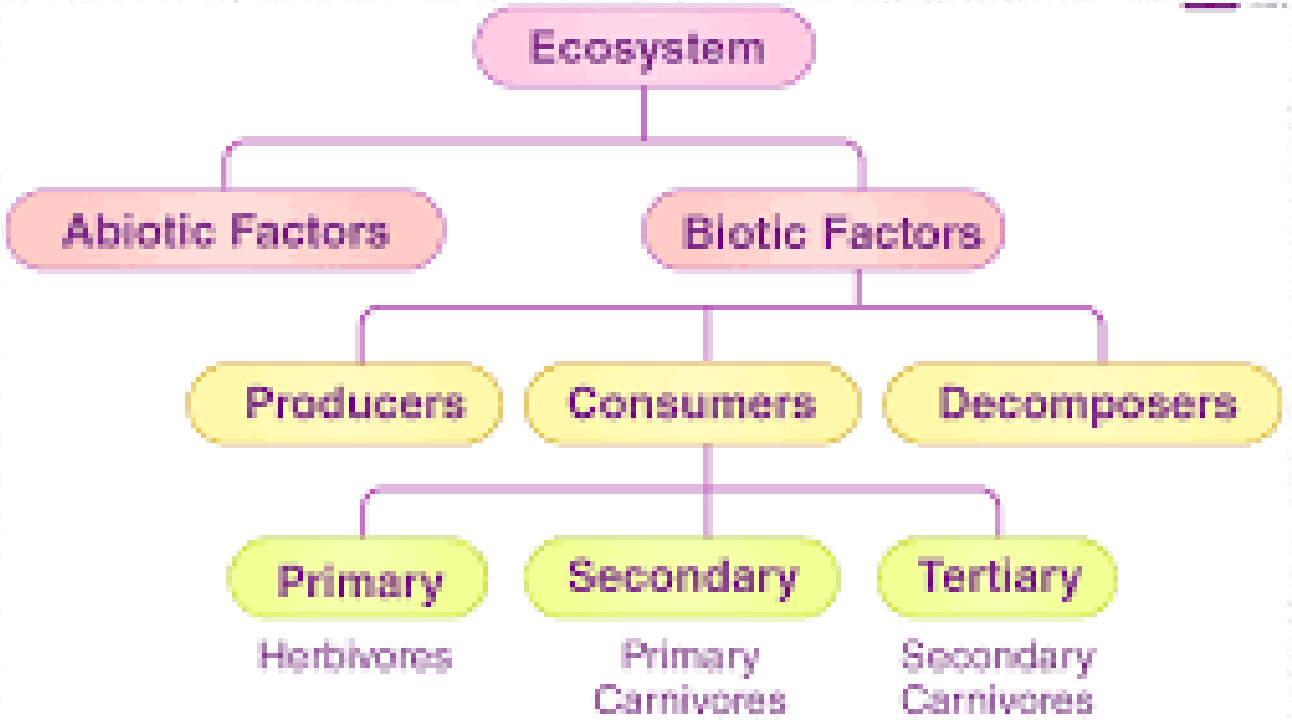
Structure of Ecosystem

- An ecosystem is a group or community composed of living and non-living things and their interactions with each other.
- Every ecosystem has two components, namely, biotic components and abiotic components.
- Biotic components refer to all living organisms in an ecology
- Abiotic components refers to the non-living things.



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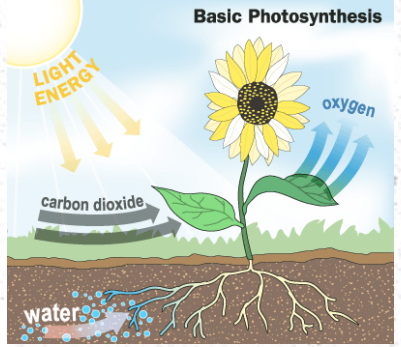

Biotic components can be classified into three categories:

Producers: These include all the autotrophs. They use light energy and synthesize food on their own, e.g. plants, green algae, etc.

Two types of Producers:

(i) Photo-autotrophes: These organisms use solar energy to produce food: Green algae, Green plants

(ii) Chemo-autotrophes: Uses chemicals (hydrogen, sulphur, iron) to produce energy, ex: methanogenic bacteria, Nitrobacter, nitrosomanas

Consumers: These include all the heterotrophs that directly or indirectly depend on producers for their food. Consumers are further categorized as herbivores, carnivores, omnivores and parasites.

Decomposers: These include saprophytes which act on dead matter and decay them for their nutrition.

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Key Functions of Ecosystem

- The main function of an ecosystem is to regulate and to promote the essential ecological processes so as to support life on earth and provide stability to the ongoing process

- There are different trophic levels of ecosystems and the work of an ecosystem is to maintain a balance between all such tropic labels

- The interaction of abiotic components with the biotic components is maintained by an ecosystem

- Ecosystem involves in carbon fixation, pollination, oxygen release, cycle nutrients, mitigate droughts and floods etc.

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Energy Flow in Ecosystems

Energy is the fundamental factor that drives all metabolic processes. Energy flow is the unidirectional flow of energy from the producer to the top consumers in Environment.

- The flow of energy through living organisms in an environment is known as energy flow.
- All living organisms can be divided into producers and consumers, and a food chain can be formed from them.
- Each nutritional level in the food chain is referred to as a trophic level.
- These food chains are then grouped into trophic pyramids to more efficiently represent the abundance of species at each trophic level.
- The energy flow is unidirectional along the food chain, with the head of an arrow showing the direction of energy flow; energy is lost as heat at each step along the way.

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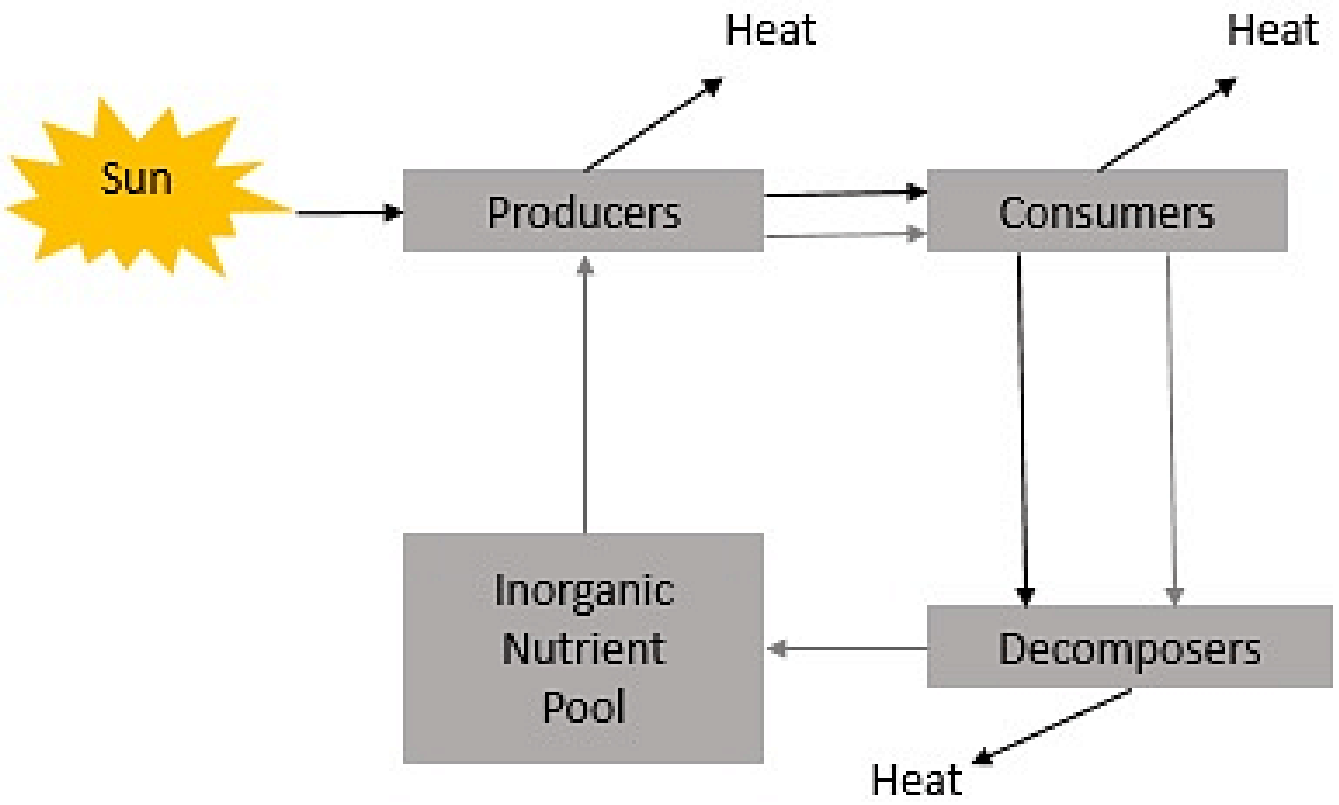

Energy Flow in an Ecosystem

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This energy flow is based on two important Laws of Thermodynamics which are as follows:

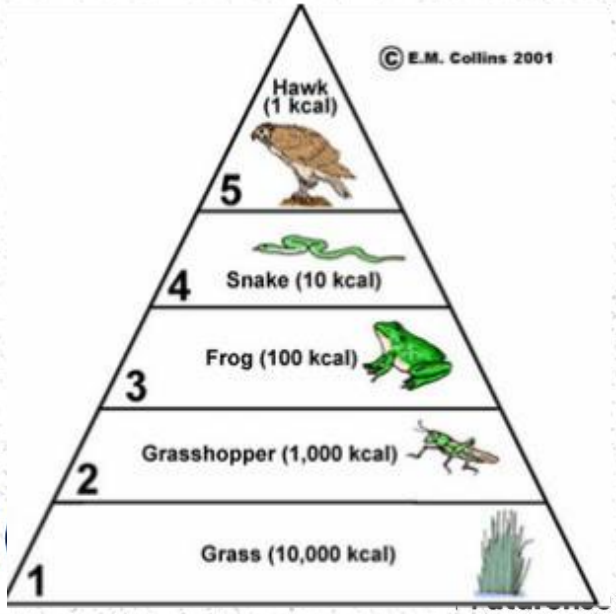
(i) First Law of thermodyanamics

It states that the amount of energy in the universe is constant. It may change from one form to another, but it can neither be created nor destroyed.
Ex: Photosynthesis, solar energy is converted into chemical energy

(ii) **Second Law of thermodyanamics**

The change of energy from one form to another takes place in such a way that a part of energy assumes waste form (heat energy). **(i-e.) Energy transformation is not 100% efficient**
Ex: Tropical food pyramid

- Much less energy is available for organisms in higher trophic levels.
- This energy is lost during metabolic processes performed by the primary consumers that are eaten.



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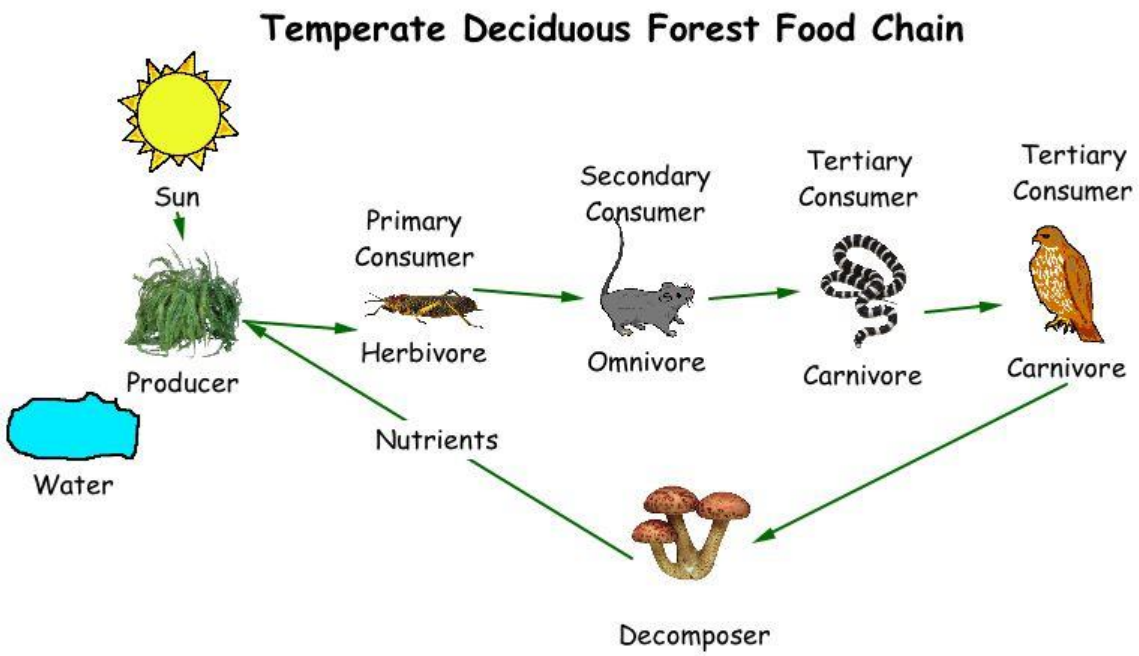
Food Chain

Transfer of food energy from green plants (producers) through a series of organisms with repeated eating and being eaten is called a food chain.



Each step in the food chain is called **trophic level**. In the above example grasses are 1st, and eagle represents the 5th trophic level. During this process of transfer of energy some energy is lost into the system as heat energy and is not available to the next trophic level. Therefore, the number of steps are limited in a chain to 4 or 5.

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Food Web

The different food chains are inter connected at various trophic level to develop a food web. For example, in grassland ecosystem, grass is consumed by the rabbit but in their absence, it may be eaten by the grazing cattle. Similarly, rat or mouse is eaten by snake but snake can be eaten by predatory birds.

In contrast to food chain, food web has several distinct characteristic.

(1) Food web are never straight.

(2) Food web is formed due to interlinking of food chains.

(3) A food web in the ecosystem brings alternate source of food.

The complex food web gives better stability to the ecosystem. Most of the animals feed on more than one kind of organism. If the availability of one particular animal is decreasing in the ecosystem, they start eating alternate animal. As a result, it gives chance to other animal to reproduce and grow in number and in addition, it gives chance to predator to survive.

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Types of Ecosystem

There are two main ecosystems:

1. **Terrestrial Ecosystems:**
a. Grassland Ecosystems
b. Forest Ecosystems
c. Desert Ecosystems
d. Cropland Ecosystems

2. **Aquatic Ecosystems:**
Aquatic ecosystems are majorly two types:
a. Freshwater/Inland Ecosystems
b. Marine/Saltwater Ecosystems

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Forest Ecosystems

A forest ecosystem is a functional unit or a system which comprises of soil, trees, insects, animals, birds, and man as its interacting units. A forest is a large and complex ecosystem and hence has greater species diversity.

Total forests cover in the world is about 33% of the world's land area. In India, the forest cover is about 22% of the total land area

Characteristics:

Forests can be found in many different climatic zones from tropical rainforests and boreal rainforests

Features:

Forests are home to variety of plants and animals

Trees in the forest ecosystem play a vital role in capturing CO2 from atmosphere and utilizing for photosynthesis which help to mitigate the climatic changes

Structure:

Forests have several layers, Top of the tallest trees is called as canpoy layer, below this there is understory layer which consists of smaller trees, floor of the forest is called as lowest layer which is covered with organic maters and fallen leaves.

Functions:

Regulating the climate, purifying the air and water, providing habitat for the wildlife, supplying timber and forest products are the key functions of forest.

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Aquatic Ecosystems

Aquatic ecosystems are those that are characterized by the presence of water. It include ocean, rivers, lakes and wetlands.

Types:

Two types mainly fresh water and marine

Characteristics:

Aquatic ecosystems are characterized by their unique physical and chemical properties, including the presence of dissolved gases, salinity and temperature.

Features:

Aquatic ecosystems have a range of physical features, such as water depth, flow rate, and substance types. These features play an important role in determining the types of plants and animals that can live in ecosystem

Structure:

Aquatic ecosystems have a complex ecological structure that includes produces, consumers and decomposers. Producers include plants and algae, which uses sunlight to produce energy through photosynthesis. Consumers include fishes, mammals. Decomposers include bacteria and fungi, which break down organic matter and recycle nutrients.

Functions:

Water filtration, nutrient cycling, climate regulation, source of food are the key functions of aquatic ecosystems.

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Grass Land Ecosystems

Grass land ecosystems are dominated by grasses and other herbaceous plants, with few or no trees
It can be found in many different climatic zones from tropical area to arctic regions

Characteristics

Grass land ecosystems are characterized by open spaces with few or no trees
Grasses are the dominant vegetation, including shrubs and other herbaceous plants

Features:

Grasslands are home to a wide variety of plant and animal species, including mammals, birds, reptiles, and insects. The soil in grass lands are rich in organic matter and helps the plant to grow efficiently.

Structure:

Grasslands are typically divided into two types Tall grass prairies and short grass prairies.
Tall grass prairies have grasses that can grow up to six feet tall, while short grass prairies have grasses that are only a few inches tall

Functions:

Grasslands also provide important area for livestock and other animals and can be used for agricultural production. Grasses in the grassland ecosystem play an important role in soil stabilization, erosion control and carbon sequestration.

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Summary

Outcomes:

a. Understanding about the various components of environment and ecosystem
b. Knowledge about various components of environment and ecosystem
c. Learning about Energy flow, food chain and food web in the ecosystem

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